

UTQIAGVIK, AK, USA

WMO: 700260

Lat: 71.283N

Lon: 156.782W

Elev: 31

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27502

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-32.5	-34.8	0.8	-28.6	-33.1	0.8	-27.5	35.8	-3.1	32.6	-1.8	7.3	200	1.104

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 58.2	(d) 53.0	(e) 54.1	(f) 50.5	(g) 50.5	(h) 47.5	(i) 53.4	(j) 57.1	(k) 50.7	(l) 54.0	(m) 47.9	(n) 50.5	(o) 11.1	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.7	55.0	55.7	48.1	49.8	52.7	45.7	45.4	49.6	22.2	57.4	20.6	54.2	19.1	50.4	61.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.1	27.3	24.8	-41.3	66.4	6.1	4.4	-45.7	69.6	-49.3	72.2	-52.7	74.6	-57.1	77.9
			-33.5	57.9	3.3	2.5	-35.9	59.7	-37.8	61.2	-39.7	62.6	-42.1	64.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	-11.1	-10.7	-10.3	4.0	22.6	35.8	41.8	40.0	34.2	21.4	6.2	-5.7
	DBStd	22.22	11.91	13.18	11.66	10.73	8.61	5.65	6.11	5.86	5.65	8.91	10.88	10.98
	HDD50	13102	1893	1700	1868	1379	851	427	266	318	473	886	1315	1726
	HDD65	18558	2358	2120	2333	1829	1316	876	720	776	923	1351	1765	2191
	CDD50	18	0	0	0	0	0	1	10	7	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.7	12.8	13.4	12.1	12.0	12.3	12.1	12.3	12.2	12.6	14.1	13.8	12.9
Precipitation	PrecAvg	5.0	0.1	0.2	0.2	0.2	0.2	0.4	0.9	1.1	0.8	0.5	0.3	0.2
	PrecMax	8.4	0.6	0.8	0.9	0.7	0.9	1.4	2.0	2.6	1.9	1.5	0.8	1.1
	PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.2	0.2	0.1	0.0
	PrecStd	1.4	0.2	0.2	0.2	0.2	0.2	0.3	0.5	0.6	0.4	0.3	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.7	27.9	27.3	34.6	40.2	58.9	64.9	62.7	54.6	38.0	33.0	28.2
		MCWB	27.5	27.1	26.8	33.3	37.9	51.9	56.4	55.5	49.7	36.2	31.7	27.3
	2%	DB	20.4	21.8	20.2	28.1	36.2	52.2	59.4	56.9	48.5	35.3	29.9	21.5
		MCWB	19.5	21.2	19.7	27.0	34.6	47.4	53.8	52.8	46.2	33.7	28.4	20.5
	5%	DB	10.0	15.1	13.2	23.7	34.4	47.9	55.1	52.9	45.0	33.4	26.4	16.1
		MCWB	9.1	14.3	12.3	22.9	33.1	44.5	51.1	50.3	43.6	31.9	25.0	15.3
	10%	DB	4.3	8.0	6.3	19.1	33.0	44.0	51.5	49.1	41.5	31.6	22.8	10.4
		MCWB	3.8	6.9	5.6	18.5	32.0	41.4	48.5	47.0	40.4	30.3	21.6	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	27.4	27.0	33.2	37.9	52.9	57.8	56.8	50.2	36.8	31.8	27.4
		MCDB	28.7	27.9	27.5	34.4	39.8	58.1	63.5	60.8	54.1	37.7	32.4	28.1
	2%	WB	20.5	21.5	19.7	27.3	34.8	47.8	54.2	53.2	46.5	33.8	28.4	20.7
		MCDB	21.4	22.1	20.1	28.1	35.9	51.9	58.7	56.2	48.3	34.9	29.5	21.5
	5%	WB	9.9	14.9	12.9	22.9	33.4	44.7	51.3	50.3	43.8	32.1	25.1	15.2
		MCDB	10.6	15.6	13.6	23.7	34.2	47.8	54.9	53.0	45.0	33.2	26.3	16.0
	10%	WB	4.2	8.0	6.0	18.6	32.2	41.6	48.6	47.2	40.7	30.4	21.4	9.8
		MCDB	4.6	8.9	6.6	19.0	32.9	44.0	51.5	49.1	41.6	31.5	22.5	10.6
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.1	12.1	7.7	8.9	11.1	8.3	6.2	7.7	10.2	10.9
		MCDBR	18.3	19.6	17.6	12.9	7.8	17.4	18.7	16.3	12.1	7.4	12.8	15.7
		MCWBR	18.0	19.4	17.4	12.4	6.7	13.5	13.9	12.1	9.6	7.1	12.4	15.5
	5% WB	MCDBR	18.7	19.8	18.0	13.0	7.5	17.2	18.3	15.8	11.9	7.3	13.4	15.6
		MCWBR	18.5	19.6	17.8	12.6	6.7	13.5	14.0	12.0	9.7	7.1	13.0	15.4
Clear-Sky Solar Irradiance	taub	N/A	0.203	0.245	0.305	0.315	0.313	0.334	0.330	0.283	0.202	N/A	N/A	
	taud	N/A	1.973	2.156	2.053	2.135	2.314	2.422	2.519	2.555	2.134	N/A	N/A	
	Ebn at Noon	0	148	239	254	270	277	266	253	234	151	0	0	
	Edh at Noon	0	11	22	36	37	33	29	23	16	9	0	0	
All-Sky Solar Radiation	RadAvg	0	88	583	1298	1765	2041	1604	888	420	104	0	0	
	RadStd	0	10	34	84	144	104	178	89	35	14	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.56	+4.36	+2.48	N/A	N/A	N/A	-772	-774	N/A	N/A
(47) Regional (3 neighbors)		+2.63	+4.82	+2.48	N/A	N/A	N/A	-869	-873	N/A	N/A

Nomenclature: See separate page